

Computational Astrophysics Exercises — Assignments Set 7

This is the only assignments set of our lecture this term that does not include a programming task (so far 😊). Instead, these exercises can be solved by pen and paper. However, since the lecture is called `_Computational_ Astrophysics`, you are highly encouraged to use a computer to solve them. While a LLM like chatgpt or claude can probably do this (feel free to try and compare to your results, see the disclaimer below), I want you to use an open source computer algebra system like SymPy or the symbolic package for octave or if need be, a commercial software package like Mathematica (free trial) or the symbolic toolbox in MATLAB (free trial or maybe via the university license?) to solve the exercises.

Numerical treatment of ordinary differential equations — Cont'd

Exercise 1: Stability of selected integration schemes

Using the ansatz by Dahlquist¹ with testequation $y'(x) = \lambda y(x)$, derive the area of stability² for the following integration schemes

(i) Explicit Euler:

$$y_{n+1} = y_n + hf(x_n, y_n). \quad (1)$$

(ii) Implicit Euler:

$$y_{n+1} = y_n + hf(x_n, y_{n+1}). \quad (2)$$

(iii) Heun's method:

$$y_{n+1} = y_n + \frac{h}{2}(f(x_n, y_n) + f(x_{n+1}, y_n + hf(x_n, y_n))). \quad (3)$$

(0b100 points)

Astrophysical Fluid Dynamics

Exercise 2: Euler equations in hyperbolic form

Consider the 1D Euler equations with no external forces, with density $\rho(x, t)$, velocity $u(x, t)$ and specific total energy $\epsilon_{\text{tot}} = \epsilon + \frac{u^2}{2}$, which describe the flow of an ideal gas with pressure $p = (\gamma - 1)\rho\epsilon$, where γ is the ratio between specific heat capacities at constant pressure and volume, respectively, and ϵ the specific internal energy.

¹https://en.wikipedia.org/wiki/Germund_Dahlquist

²See lecture slides 47ff in 05_ODEs.pdf on ILIAS.

- (i) With the use of the vector of the conserved quantities

$$\mathbf{u} = (u_1, u_2, u_3)^T = (\varrho, \varrho u, \varrho \epsilon_{\text{tot}})^T, \quad (4)$$

rewrite the Euler equations (in Eulerian representation) in the following form

$$\frac{\partial \mathbf{u}}{\partial t} + \frac{\partial \mathbf{f}}{\partial x} = 0, \quad (5)$$

by writing the components of the vector of fluxes

$$\mathbf{f} = (\varrho u, \varrho u^2 + p, (\varrho \epsilon_{\text{tot}} + p)u)^T \quad (6)$$

in terms of the conserved quantities (u_1, u_2, u_3) . Then, transform the system to the following form

$$\frac{\partial \mathbf{u}}{\partial t} + A \frac{\partial \mathbf{u}}{\partial x} = 0, \quad (7)$$

where A denotes the Jacobian $\partial \mathbf{f} / \partial \mathbf{u}$.

Calculate the components a_{ij} of A .

- (ii) Again, consider the 1D Euler equations without any external forces. The vector of the primitive variables is

$$\mathbf{w} = (\varrho, u, p)^T.$$

- (a) Re-write the Euler equations in the following form

$$\frac{\partial \mathbf{w}}{\partial t} + C \frac{\partial \mathbf{w}}{\partial x} = 0, \quad (8)$$

and calculate the elements of the matrix C .

- (b) A system of PDEs is called hyperbolic if the matrix
- C
- can be diagonalisable with real eigenvalues. Show, that
- C
- can be diagonalised and calculate the eigenvalues
- λ_i
- of
- C
- .

(0b110 points)

A note on the use of AI 🤖:

The use of AI tools (e.g. ChatGPT, Claude, Copilot) is permitted for these exercises, provided that (1) any use is clearly documented in your submission (which tool, for what purpose), and (2) you are able to explain all submitted solutions in full — on request, during the exercise class or otherwise. Submissions that cannot be explained by the submitter(s) (that includes all members of the submitting team) will be graded as failed, regardless of whether AI was used.

in total

(0b1010 points)

Deadline:

Fri, 26 June at noon  via email (including code) to christoph.schaefer@uni-tuebingen.de.